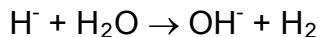


Section A

For each question there are four possible answers, **A**, **B**, **C**, and **D**. Choose the one you consider to be correct.

- 1 Group I and Group II ionic hydrides react with water.



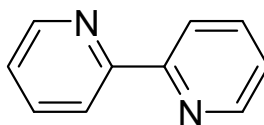
In an experiment, 1 g of a sample of an ionic hydride is dissolved in excess H_2O . The resulting solution required 24.0 cm^3 of a 2.0 mol dm^{-3} HCl solution for complete neutralization.

What is the formula of the hydride?

- A** LiH
B NaH
C MgH_2
D CaH_2
- 2 Which one of the following pairs contains substances with similar shape?
- A** CO_2 and I_3^-
B NH_3 and NO_3^-
C BeCl_2 and SnCl_2
D BCl_3 and PCl_3
- 3 Carbon dioxide is a gas at room temperature while silicon dioxide is a solid because
- A** van der Waals' forces are much weaker than covalent bonds.
B carbon-oxygen bonds are less polar than silicon-oxygen bonds.
C the relative molecular mass of carbon dioxide is less than that of silicon dioxide.
D carbon dioxide contains double covalent bonds while silicon dioxide contains single covalent bonds.
- 4 Which one of the following pairs of graphs share the same general shape according to ideal gas law for a fixed mass of gas?
- A** V against T (at constant P) and PV against V (at constant T)
B V against $1/P$ (at constant T) and P against $1/T$ (at constant V)
C T against $1/P$ (at constant V) and $1/\text{density}$ against T (at constant P)
D PV against P (at constant T) and V/T against T (at constant P)

- 5 Chromium has a proton number of 24.

What is the electronic configuration of the chromium which can form the complex, $\text{Cr}(\text{CO})_4(\text{bpy})$, where bpy is a bidentate ligand?



bpy = bipyridine

- A $[\text{Ar}]3\text{d}^44\text{s}^2$
 B $[\text{Ar}]3\text{d}^54\text{s}^1$
 C $[\text{Ar}]3\text{d}^44\text{s}^1$
 D $[\text{Ar}]3\text{d}^5$
- 6 The standard enthalpy change of formation of carbon dioxide and water are -394 kJ mol^{-1} and -286 kJ mol^{-1} respectively.

What is the standard enthalpy change of formation of propyne, C_3H_4 , given that the enthalpy change of combustion of propyne is $-1938 \text{ kJ mol}^{-1}$?

- A $+184 \text{ kJ mol}^{-1}$
 B -184 kJ mol^{-1}
 C $+1258 \text{ kJ mol}^{-1}$
 D $-1258 \text{ kJ mol}^{-1}$
- 7 For which one of the following is the sign of ΔS **incorrect**?

	Reactions	ΔS
A	$\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightarrow 2\text{NH}_3(\text{g})$	positive
B	$\text{HCl}(\text{aq}) + \text{NaHCO}_3(\text{aq}) \rightarrow \text{NaCl}(\text{aq}) + \text{CO}_2(\text{g}) + \text{H}_2\text{O}(\text{l})$	positive
C	$\text{H}_2\text{O}(\text{l}) \rightarrow \text{H}_2\text{O}(\text{s})$	negative
D	$\text{NH}_4\text{NO}_3(\text{s}) \rightarrow \text{NH}_4^+(\text{aq}) + \text{NO}_3^-(\text{aq})$	positive

- 8 Ammonia usually reacts as a base.

In which reaction does it behave as an acid?

- A $\text{HCl}(\text{g}) + \text{NH}_3(\text{g}) \rightarrow \text{NH}_4\text{Cl}(\text{s})$
 B $\text{Cu}^{2+}(\text{aq}) + 4\text{NH}_3(\text{aq}) \rightarrow [\text{Cu}(\text{NH}_3)_4]^{2+}(\text{aq})$
 C $2\text{Na}(\text{l}) + 2\text{NH}_3(\text{g}) \rightarrow 2\text{NaNH}_2(\text{s}) + \text{H}_2(\text{g})$
 D $\text{CH}_3\text{Cl}(\text{alc}) + \text{NH}_3(\text{alc}) \rightarrow \text{CH}_3\text{NH}_2(\text{alc}) + \text{HCl}(\text{alc})$

- 9 $\text{Cu}_3(\text{PO}_4)_2$ is a sparingly soluble salt.

If its solubility product is s , what is the concentration of $\text{Cu}^{2+}(\text{aq})$ at equilibrium?

- A $(s/108)^{1/5}$
 B $(9s/4)^{1/5}$
 C $(s/6)^{1/2}$
 D $(3s/2)^{1/2}$

- 10 An experiment was carried out to investigate the initial rate of reaction between ammonium peroxodisulphate, $(\text{NH}_4)_2\text{S}_2\text{O}_8$, an oxidising agent, and potassium iodide, KI. The initial concentrations of the $(\text{NH}_4)_2\text{S}_2\text{O}_8$ and KI solutions in the mixture together with the time taken for the mixture to darken for the various experimental runs are given below.

Initial concentration of $(\text{NH}_4)_2\text{S}_2\text{O}_8$ /mol dm ⁻³	Initial concentration of KI /mol dm ⁻³	Time taken to darken /s
0.10	0.20	35
0.05	0.20	70
0.10	0.067	105
0.02	0.75	?

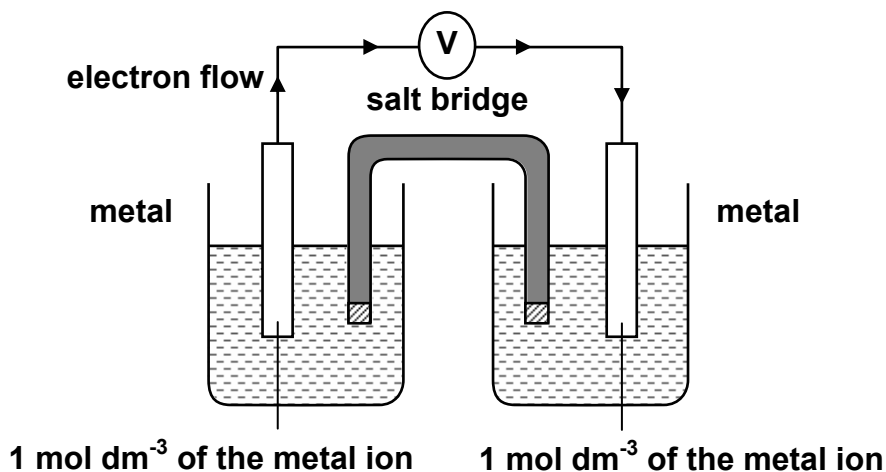
What is the expected time taken (in seconds) to darken when the experiment is repeated using initial concentrations of $(\text{NH}_4)_2\text{S}_2\text{O}_8$ and KI to be 0.02 mol dm⁻³ and 0.75 mol dm⁻³ respectively?

- A 40
 B 47
 C 60
 D 72
- 11 Which one of the following statements best explains why a small increase in temperature leads to a significant increase in the rate of a gaseous reaction?
- A The frequency of collisions between the molecules is greater at a higher temperature.
 B The activation energy of the reaction is lower when the gases are at a higher temperature.
 C The average kinetic energy of the molecules is slightly greater at a higher temperature.
 D The frequency of effective collisions between molecules with kinetic energy greater than the activation energy is greater at a higher temperature.

12 Use of the Data Booklet is relevant to this question.

The half-cells for four metals: Mg, **A**, **B** and **C** were in turn connected in pairs and the potential difference was recorded.

Temperature = 25°C



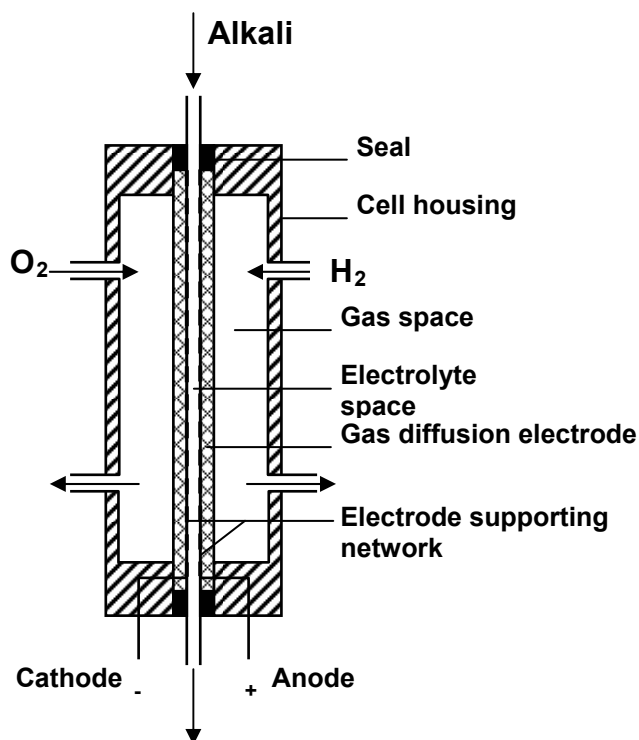
The results obtained are as shown in the table below.

Positive electrode	Negative electrode	e.m.f /V
A	Mg	+2.10
B	Mg	+2.72
Mg	C	+0.33

Rank the four metals in the order of decreasing reducing power.

- A** **B** > **A** > Mg > **C**
B **A** > **B** > Mg > **C**
C **C** > Mg > **A** > **B**
D **C** > Mg > **B** > **A**

- 13 This is a diagram of an electric cell, **X**.



Which one of the following statements is **not** true about electric cell **X**?

- A One difference between electric cell **X** and ordinary batteries is that reactants are constantly supplied and products removed.
 - B Electric cell **X** does not cause pollution but it is very expensive.
 - C Electric cell **X** can work indefinitely as long as there is a supply of H_2 and O_2 .
 - D The alkali needs to be replenished as hydroxide ions are being used up.
- 14 **W**, **X**, **Y** and **Z** are four elements in Period 3.

W is a conductor of electricity and does not react with water. **Z** forms a chloride, ZCl_5 which reacts with water and dissolves in organic solvent. **Y** forms a solid oxide YO_2 . XOH is a strong alkali.

Arrange the above elements in order of increasing proton number.

- A **X**, **W**, **Y**, **Z**
- B **X**, **Y**, **W**, **Z**
- C **W**, **Y**, **X**, **Z**
- D **W**, **X**, **Y**, **Z**

- 16** The table below shows the results of experiments in which the three potassium halides were added separately to concentrated H_2SO_4 .

KX(s)	KY(s)	KZ(s)
Orange brown fumes formed	Purple vapour formed	White fumes produced

In which sequence is the solubility of the silver halides in aqueous ammonia arranged in **increasing** order?

- 17 Platinum(IV) chloride combines with ammonia to form compound **Z** in which the co-ordination number of platinum is 6. When solution **Z** is treated with an excess aqueous silver nitrate, only one-half of the total chloride present is precipitated as AgCl.

What is the formula of compound **Z**?

- | | | | |
|----------|---------------------------------------|----------|---------------------------------------|
| A | $\text{Pt}(\text{NH}_3)_6\text{Cl}_4$ | B | $\text{Pt}(\text{NH}_3)_5\text{Cl}_4$ |
| C | $\text{Pt}(\text{NH}_3)_3\text{Cl}_4$ | D | $\text{Pt}(\text{NH}_3)_4\text{Cl}_4$ |

- 18 The table shows the electronic configuration of the three d-block elements in the Periodic Table.

Element	Electronic configuration
X	$[\text{Ar}]3d^74s^2$
Y	$[\text{Ar}]3d^84s^2$
Z	$[\text{Ar}]3d^{10}4s^1$

Which one of the following statement is **incorrect**?

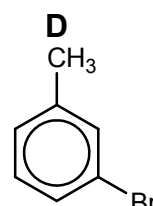
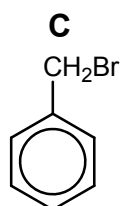
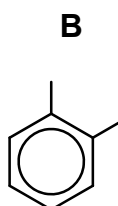
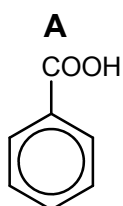
- A** The electronic configuration of central metal ion for $[\text{Y}(\text{CN})_6]^{4-}$ is $[\text{Ar}]3d^8$.
- B** Upon reduction from $\text{ZCl}_2(\text{aq})$ to $[\text{ZCl}_2]^{-}(\text{aq})$, the solution turned colourless.
- C** The $E^\ominus$ value of $\text{X}^{3+}/\text{X}^{2+}$ is less positive than that of $\text{Z}^{3+}/\text{Z}^{2+}$.
- D** **X** is likely to exist as $\text{K}_2\text{X}_2\text{O}_7$.
- 19 The hydride ion, H^- , is a strong reducing agent, a good nucleophile and a good base.

Which one of the following conversions would the hydride ion **not** be expected to bring about?

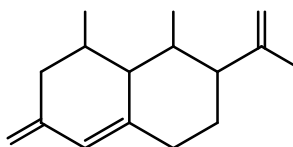
- A** $\text{C}_2\text{H}_5\text{Br}$ to C_2H_6
- B** CH_3CH_3 to $\text{CH}_2=\text{CH}_2$
- C** CH_3CHO to $\text{CH}_3\text{CH}_2\text{OH}$
- D** $\text{CH}_3\text{CO}_2\text{H}$ to CH_3CO_2^-
- 20 2,2-dimethylpropane reacts with chlorine gas in the presence of ultraviolet light to give a mixture of products.

Which one of the following statements is correct regarding this reaction?

- A** Only one mono-substituted product is formed.
- B** Only the propagation step involves C-Cl bond formation.
- C** Only the initiation step of the mechanism involves homolytic fission.
- D** Both propagation and termination steps produce hydrogen chloride.
- 21 Which one of the following **cannot** be made directly from methylbenzene?



- 22** Nootkatone is one of the organic compounds found in grapefruit. Its structure is as shown below.



Which statement is correct about nootkatone?

- A** It forms a pale yellow precipitate with alkaline I₂(aq).
B It reacts with Tollen's reagent to give a silver mirror.
C It reacts with potassium cyanide in ethanol to form a nitrile compound.
D It reacts with hot acidified potassium manganate(VII) to form one organic product.
- 23** An organic compound **X** undergoes the following reactions:
- (i) It decolourises a solution of bromine in tetrachloromethane.
 - (ii) It reacts with phosphorus pentachloride giving copious white fumes of HCl.
 - (iii) It reacts with hot alkali to give a compound with two alcohol functional groups.

Which compound could be **X**?

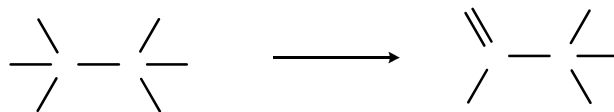
- A** HOH₂CCH=CHCH=CHCH₂Cl
B Cl₂CHCH=CHCH₂COOH
C BrCH₂CH₂CHClCH₂COCl
D ClCH₂CH₂CH=CHCH(Cl)CH₂OH
- 24** Which process will **not** give a good yield of CH₃CO(CH₂)₃CO₂H?

- A**

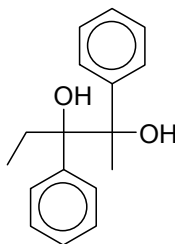
$$\text{1-chloro-2-methylcyclopentane} \xrightarrow[\text{heat}]{\text{KOH in ethanol}} \text{alkene} \xrightarrow{\text{Hot KMnO}_4(\text{aq}), \text{H}_3\text{O}^+} \text{CH}_3\text{CO}(\text{CH}_2)_3\text{CO}_2\text{H}$$
- B**
$$\text{H}_2\text{NCO}(\text{CH}_2)_3\text{CHClCH}_3 \xrightarrow[\text{heat}]{\text{KOH(aq)}} \text{alkene} \xrightarrow{\text{Hot KMnO}_4(\text{aq}), \text{H}_3\text{O}^+} \text{CH}_3\text{CO}(\text{CH}_2)_3\text{CO}_2\text{H}$$
- C**
$$\text{CH}_3\text{CH}(\text{OH})(\text{CH}_2)_3\text{CH}_2\text{Cl} \xrightarrow[\text{Heat}]{\text{NaOH(alcohol)}} \text{alkene} \xrightarrow{\text{Hot KMnO}_4(\text{aq}), \text{H}_3\text{O}^+} \text{CH}_3\text{CO}(\text{CH}_2)_3\text{CO}_2\text{H}$$
- D**

$$\text{1-methylcycloheptene} \xrightarrow{\text{Hot KMnO}_4(\text{aq}), \text{H}_3\text{O}^+} \text{CH}_3\text{CO}(\text{CH}_2)_3\text{CO}_2\text{H}$$

- 25 The pinacol rearrangement involves the reaction of a diol in acidic conditions to form a carbonyl compound as shown below.

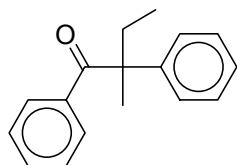


Which one of the following structural formulae is **not** a product of the pinacol rearrangement of 2,3-diphenylpentane-2,3-diol with the structure as shown below?

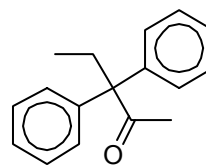


2,3-diphenylpentane-2,3-diol

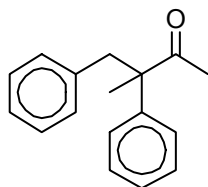
A



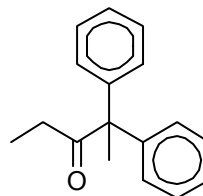
C



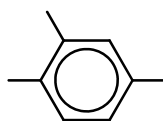
B



D



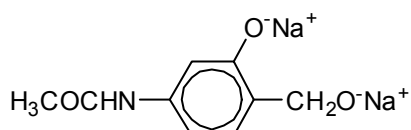
- 26 Compound **X** is boiled under reflux with an excess of aqueous sodium hydroxide.



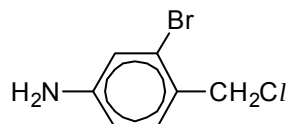
Compound **X**

What is the main product?

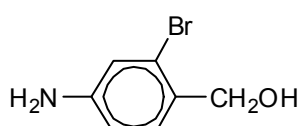
A



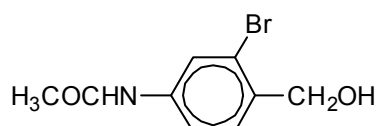
C



B



D

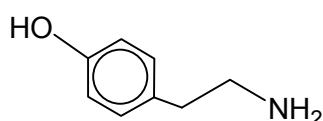


- 27 An ester **Y** is boiled with aqueous hydrochloric acid and the resulting mixture is cooled. The final products include a compound $C_4H_8O_2$ which gives a pale yellow precipitate with alkaline aqueous iodine.

Which formula could represent **Y**?

- A** $CH_3OCOCH_2CH_2CH_3$
B $C_6H_5CO_2CH_2CH_2CH_2CHO$
C $CH_3CO_2CH_2CH_2COCH_3$
D $C_6H_5CO_2CH(CH_3)CH_2CH_2OH$

- 28 Tyramine is an alkaloid found, among other places, in mistletoe and ripe cheese.



Tyramine

How many moles of ethanoyl chloride react **completely** with one mole of tyramine?

- A** 0 **B** 1 **C** 2 **D** 3

- 29 Which sequence correctly shows the compounds, in aqueous solutions of equal concentration, in order of **increasing** pH?

- A** $C_2H_5OH < C_6H_5COCl < CH_3CH(NH_2)CO_2H < CH_3CH_2NH_2$
B $CH_3CH_2OH < CH_3CH_2NH_3^+Cl^- < C_6H_5NH_2 < CH_3CH_2NH_2$
C $C_6H_5OH < C_6H_5NH_2 < CH_3CONH_2 < CH_3CH_2NH_3^+Cl^-$
D $CH_3COCl < C_6H_5CO_2H < CH_3CONH_2 < C_6H_5NH_2$

- 30 A liquid **P** is sparingly soluble in water but dissolves readily in cold hydrochloric acid. Evaporation of this solution yields a crystalline solid.

What is the identity of **P**?

- A** $C_6H_5NH_2$
B $C_6H_5CH_3$
C $CH_3(CH_2)_{10}CONH_2$
D $HO_2CCH(CH_3)NH_2$

Section B

For each of the questions in this section, one or more of the three numbered statements **1** to **3** may be correct.

Decide whether each of the statements is or is not correct (you may find it helpful to put a tick against the statements which you consider to be correct).

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

31 Consider the following enthalpy changes in kJ mol^{-1} :



Which statements are **incorrect**?

- 1** **R** is equal to the first ionisation energy of Mg.
- 2** **(Q – S – R)** is greater than **P**.
- 3** **T** is **not** equal to the sum of **R** and **S**.

32 Tropaeolin OOO indicator has a pH range of 7.6 to 9.0. Its colour is yellow at $\text{pH} < 7.6$ and rose-red at $\text{pH} > 9.0$.

Which of the following statements about Tropaeolin OOO indicator are correct?

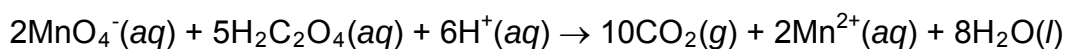
- 1** The indicator is yellow when 50 cm^3 of 0.20 mol dm^{-3} of aqueous NH_3 is mixed with 50 cm^3 of 0.85 mol dm^{-3} aqueous NH_4Cl (pK_a of NH_4^+ is 9.25).
- 2** It is suitable for titration of ethanoic acid against sodium hydroxide.
- 3** The indicator is rose-red in a solution of 0.01 mol dm^{-3} of methylamine ($K_b = 4.37 \times 10^{-4} \text{ mol dm}^{-3}$).

- 33** The discharge at the cathode of 1.00 mol of rhodium ions from an aqueous solution of a rhodium salt requires 2.90×10^5 C of electricity.

Which of the following conclusions can be drawn from these observations?

- 1 The rhodium ions are positively charged.
- 2 The magnitude of the charge on the rhodium ions is three times the charge of an electron.
- 3 Rhodium is a transition element.

- 34** When a dilute, aqueous solution of potassium manganate(VII) is run from a burette into a flask containing dilute, aqueous oxalic acid ($\text{H}_2\text{C}_2\text{O}_4$) and dilute sulfuric acid, the rate of reaction suddenly increases considerably as more potassium manganate(VII) is added.



Which of the following statements explain the sudden increase in the rate of reaction?

- 1 The manganese(II) ions produced catalyse the reaction.
- 2 The dissolution of CO_2 in water produces H^+ ions and pushes the reaction forward.
- 3 Sulfuric acid acts as a dehydrating agent and removes water to push the reaction forward.

- 35** *Use of the Data Booklet is relevant to this question.*

Which of the following statements are correct for the various halides?

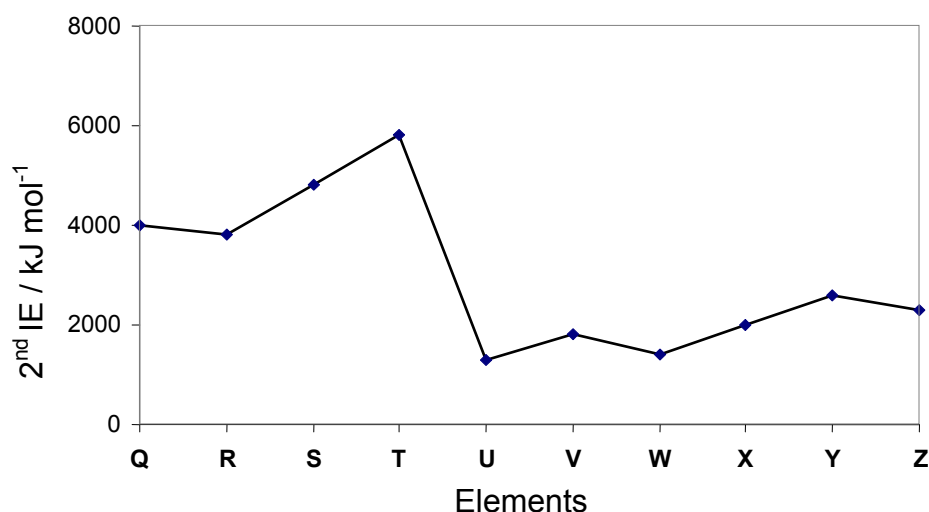
- 1 The lattice energy of magnesium chloride is numerically greater than that of calcium bromide since the interionic distance in magnesium chloride is shorter than that in calcium bromide.
- 2 The dark brown colour of solution disappears when aqueous sodium thiosulfate is added to the mixture formed by reacting concentrated sulfuric acid with solid potassium iodide.
- 3 A red liquid is produced when aqueous chlorine is added to aqueous sodium bromide.

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

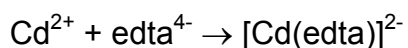
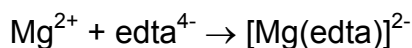
- 36** The graph below shows the variation in the **second** ionisation energies for the consecutive elements **Q** to **Z** in the Periodic Table, all with proton number below 20.



What can be deduced from the above?

- 1** **T** has a smaller atomic radius than **U**.
- 2** Effervescence is observed when a magnesium strip is dipped into the aqueous solution containing the chloride of element **W**.
- 3** When the oxides of elements **X** and **U** are mixed with water, the solution obtained is approximately neutral.

- 37** The equations below show the reactions between a solution of edta and two divalent metal ions:



where edta^{4-} has the formula of $(\text{CH}_2\text{CO}_2^-)_2\text{NCH}_2\text{CH}_2\text{N}(\text{CH}_2\text{CO}_2^-)_2$.

Which of the following statements about the above reactions are true?

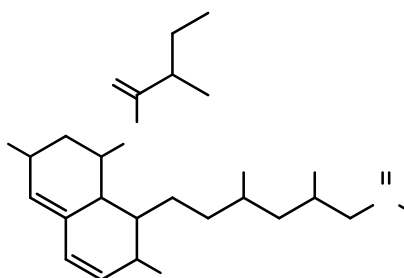
- 1** Both are ligand exchange reactions.
- 2** Both $[\text{Mg}(\text{edta})]^{2-}$ and $[\text{Cd}(\text{edta})]^{2-}$ are octahedral complexes.
- 3** The product formed from each reaction is coloured.

- 38** Compound **X** changes the colour of excess hot acidified potassium dichromate(VI) from orange to green.

Which of the following can be compound **X**?

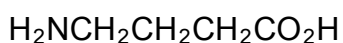
- 1 $\text{C}_6\text{H}_5\text{CH}_2\text{CHO}$
- 2 $\text{C}_6\text{H}_5\text{CH}(\text{OH})\text{CH}_3$
- 3 $\text{C}_6\text{H}_5\text{CO}_2\text{CH}_2\text{CH}_3$

- 39** The cholesterol-lowering agents called *statins*, such as pravastatin, are among the most widely prescribed drugs in the world.



Which statements about pravastatin are correct?

- 1 The pravastatin molecule has 8 chiral centres.
 - 2 When treated with an excess of hot concentrated acidified KMnO_4 , pravastatin will produce a compound which gives an orange precipitate with 2,4-dinitrophenylhydrazine.
 - 3 When 1 mole of pravastatin is treated with an excess of sodium, 2 moles of hydrogen is produced.
- 40** **GABA** is a neurotransmitter released by red algae which encourages shellfish larvae to settle on the ocean bed.



GABA

Which statements about **GABA** are correct?

- 1 It is a 2-aminocarboxylic acid.
- 2 It is soluble in water due to zwitterion formation.
- 3 It migrates to the anode of an electrolytic cell at pH 12.